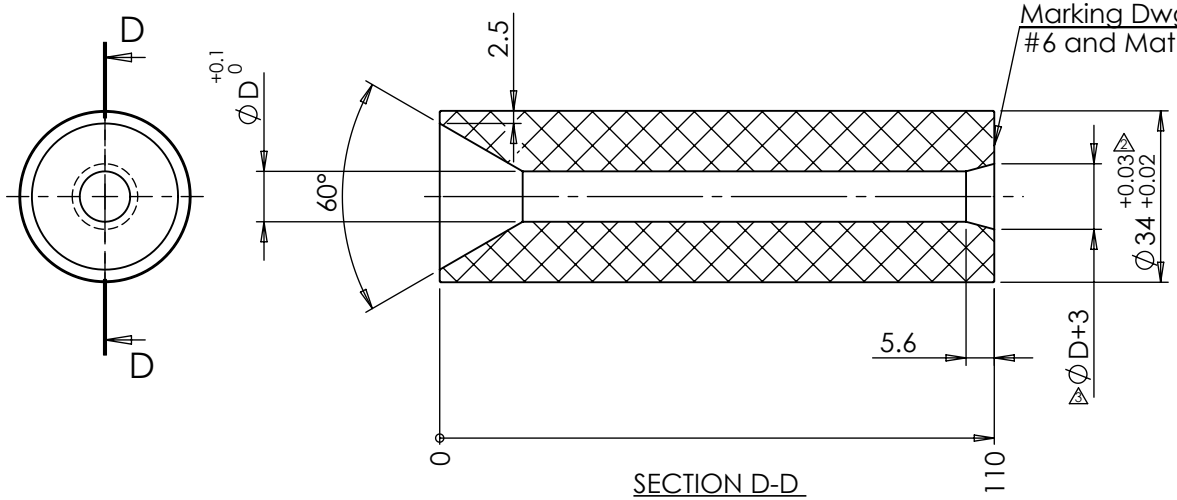
PART No.
TGM-12-390-2-xx

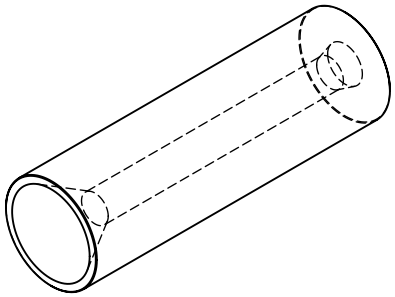
6 REV.

Rt=12

☑	Mat'l Tick by user
☐	Pom black
☐	PP Gray
☐	Peek Gray
☐	Pom white



SECTION D-D
SCALE 1 : 1.5



MinebeaMitsumi Inc.
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6/10/2569
Engineering Design center
G/M.1 & M/A. Div.

Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1
3. Vendor Ass'y TGM-12-390-1 = TGM-12-390-2 (Both Side)

Use for: Pusher

Item	D +0.1 0	For Mat'l	Check
-01	Ø10	Ø8.51 to Ø9.5	
-02	Ø12	Ø10.51 to Ø11.5	
-03	Ø7	Ø5.51 to Ø6.5	
-04	Ø8	Ø6.51 to Ø7.5	RQ.2875
-05	Ø18	Ø16.9 to 17.8	For PMA.
-06	Ø9	Ø7.51 to Ø8.5	RQ.2969
-07	Ø6	Ø5.01 to Ø5.5	RQ.118/14
-08	Ø6.5	Ø6.1 to Ø6.2	RQ.773/14
-09	Ø15	Ø14 to 14.8	RQ.267/16
-10	Ø17	Ø16 to 16.8	RQ.267/16
-11	Ø13	Ø11.6 to 12.8	RQ.271/16
-12	Ø16	Ø15 to 15.8	RQ.346/16
-13	Ø14	Ø13.2	RQ.346/16
-14	Ø11	Ø9.51 to 10.5	RQ.346/16
15	Ø20.5	Ø20	RQ.346/16

Item	D +0.1 0	For Mat'l	Check
-16	Ø18.5	Ø18	RQ/321
-17	Ø4.6	Ø4.4	Preecha Rq.304/26
-18			
-19			
-20			
-21			
-22			
-23			
-24			
-25			
-26			
-27			
-28			
-29			
-30			

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE
2	----	22 Jul'11	Preecha	Change: 35 ==> 34 etc.;
3	----	9 Jun'16	Preecha	Add: ØD + 3 etc.; Rq. 267/16
4	----	5 Sep'16	Preecha	Change: SUS ==> Pom White etc.:
5	----	7 May' 20	Preecha	Add: Item Mat'l Rq.174/20
6	----	1 Jun' 21	Preecha	Add: Item -16 (Qt'y 2)

UNLESS SPECIFIED C0.1 OR R0.1	
GENERAL TOLERANCE	
JISB0405 MACHING WORKING	
LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S 6.3S 25S 100S ~ UNCUT

M/C. BM-165 | Q'ty 2
MATERIAL See table (4) (5)
FINISH -
HEAT TREATMENT -

List Request No. -

3RD ANGLE

SCALE : | SHEET / | UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Chonknok	Preecha	Preecha	Preecha

DATE 21 Mar' 11

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

TITLE: Bush

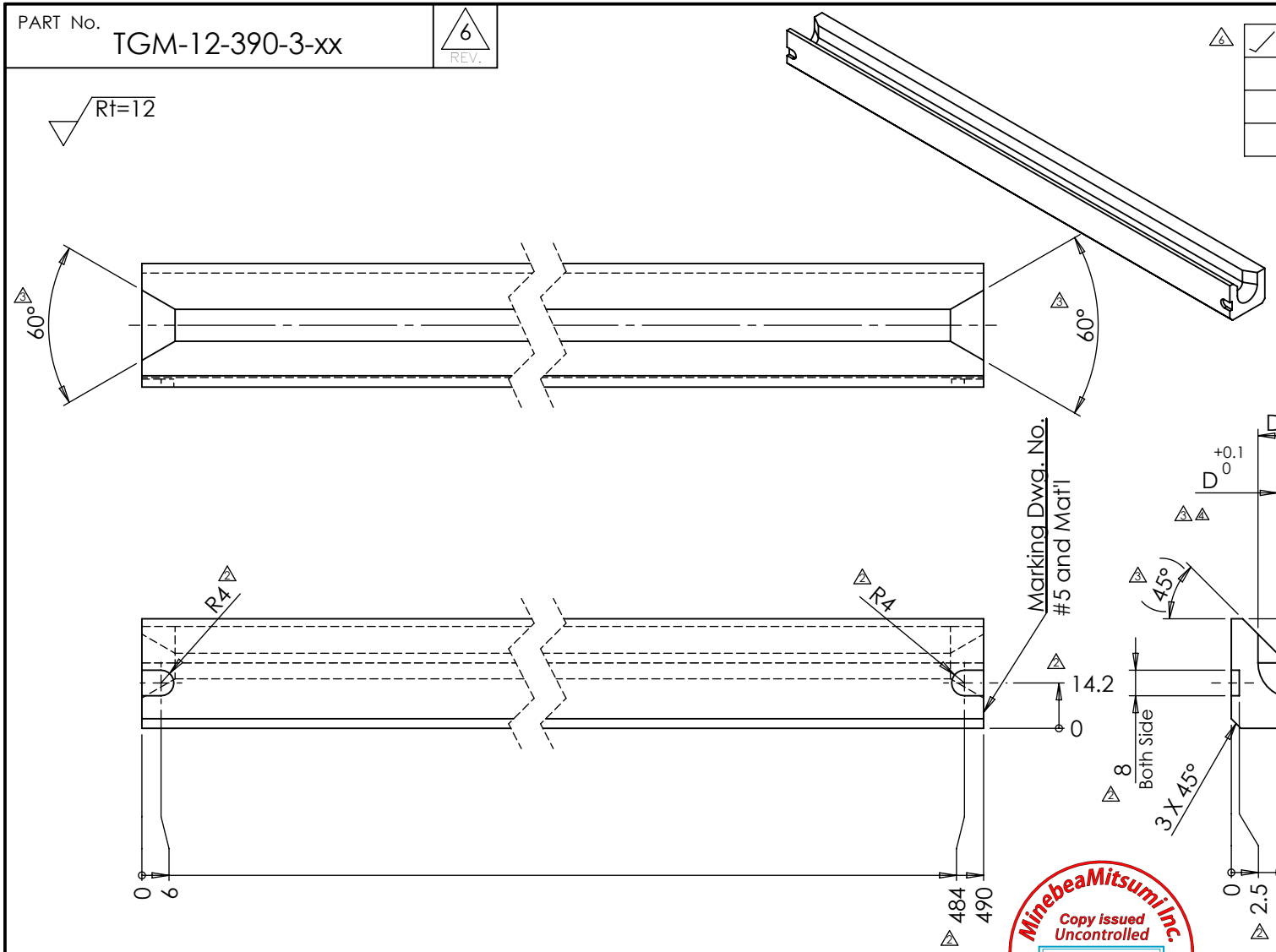
PART No. TGM-12-390-2-xx

6 REV.

PART No.
TGM-12-390-3-xx

6 REV.

Rt=12



	Mat'l Tick by user
	Pom black
	PP Gray
	Peek Gray

Item	D +0.1 0	For Mat'l	Check
-01	Ø10	Ø8.51 to Ø9.5	Preecha
-02	Ø12	Ø10.51 to Ø11.5	Preecha
-03	Ø7	Ø5.51 to Ø6.5	Preecha
-04	Ø11	Ø9.51 to Ø10.5	Preecha
-05	Ø8	Ø6.51 to Ø7.5	RQ.2875
-06	Ø18	Ø16.9 to Ø17.8	For PMA.
-07	Ø9	Ø7.51 to Ø8.5	RQ.2969
-08	Ø6	Ø5.01 to Ø5.5	RQ.118/14
-09	Ø6.5	Ø6.1 to Ø6.2	RQ.773/14
-10	Ø16	Ø15 to Ø15.8	RQ.346/16
-11	Ø15	Ø14 to Ø14.8	RQ.263/16
-12	Ø14	Ø13.2	RQ.263/16
-13	Ø18	Ø17	RQ.507/16
-14	Ø20.5	Ø20	RQ.507/16
-15	Ø18.5	Ø18	RQ.13/21

Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1

Use for: Pusher

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Engineering Design center
G/M.1 & M/A. Div.

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE
2	----	6 Feb'12	Preecha	Add: Slot etc.;
3	----	22 Feb'13	Preecha	Change: 70° ==> 45° etc.; Rq. 2836
4	----	21 Oct'13	Preecha	Add: 23.5, Delete 4 etc.; Rq. 3276
5	----	27 Jul'16	Preecha	Change: 4 ==> 2 etc.;
6	----	7 May' 20	Preecha	Add: Item Mat'l Rq.174/20

UNLESS SPECIFIED C0.1 OR R0.1	
GENERAL TOLERANCE	
JISB0405 MACHING WORKING	
LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S 6.3S 25S 100S UNCUT

M/C. BM-165 | Q'ty 2

MATERIAL See table

FINISH -

HEAT TREATMENT -

List Request No. -

3RD ANGLE

SCALE : | SHEET / | UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Chonknok	Preecha	Preecha	Preecha

DATE 21 Mar' 11

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

TITLE: G-Channel (1)

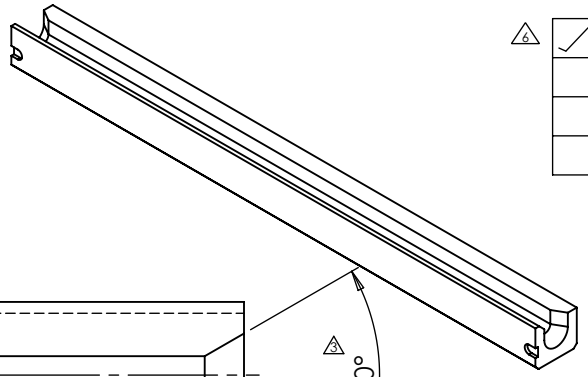
PART No. TGM-12-390-3-xx

6 REV.

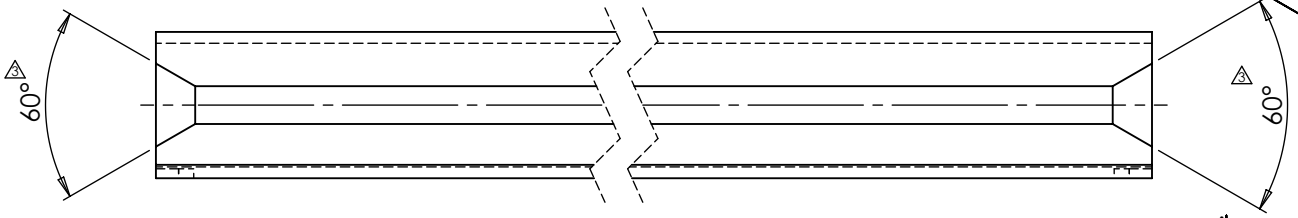
PART No.
TGM-12-390-3-xx

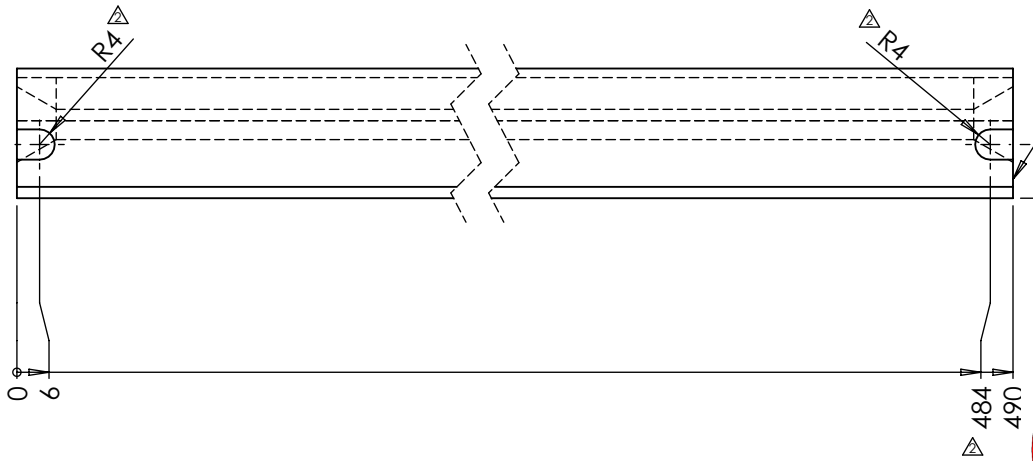
6 REV.

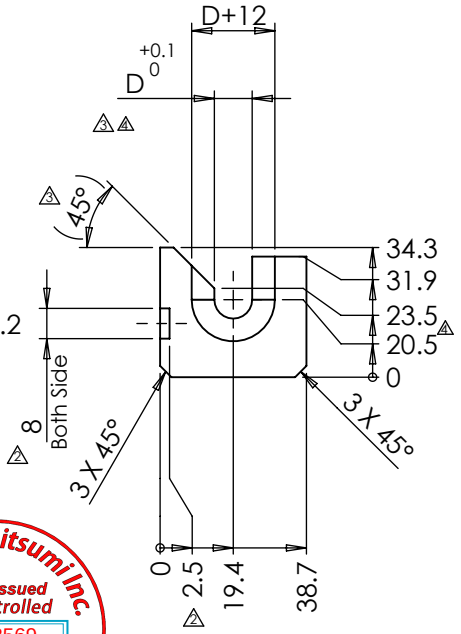
Rt=12



⚠6	Mat'l Tick by user
	Pom black
	PP Gray
	Peek Gray







MinebeaMitsumi Inc.
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Engineering
Design center
G/M.1 & M/A. Div.

Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1

Use for: Pusher

Item	D +0.1 0	For Mat'l	Check
-16	Ø4.4	Ø4.4	Preecha RQ.304/26
-17			
-18			
-19			
-20			
-21			
-22			
-23			
-24			
-25			
-26			
-27			
-28			
-29			
-30			

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE
2	----	6 Feb'12	Preecha	Add: Slot etc.;
3	----	22 Feb'13	Preecha	Change: 70° ==> 45° etc.; Rq. 2836
4	----	21 Oct'13	Preecha	Add: 23.5, Delete 4 etc.; Rq. 3276
5	----	27 Jul'16	Preecha	Change: 4 ==> 2 etc.;
6	----	7 May' 20	Preecha	Add: Item Mat'l Rq.174/20

UNLESS SPECIFIED C0.1 OR R0.1	
GENERAL TOLERANCE	
JISB0405 MACHING WORKING	
LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S 6.3S 25S 100S ~ UNCUT

M/C. BM-165 | Q'ty 2 ⚠5

MATERIAL See table ⚠6

FINISH -

HEAT TREATMENT -

List Request No. -

3RD ANGLE

SCALE : | SHEET / | UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Chonknok	Preecha	Preecha	Preecha

DATE 21 Mar' 11

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

TITLE: G-Channel (1)

PART No. TGM-12-390-3-xx

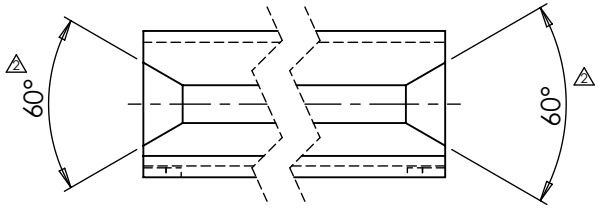
6 REV.

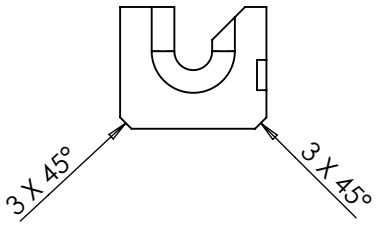
PART No.
TGM-12-390-5-xx

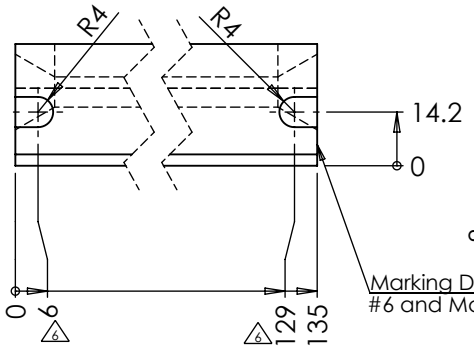
6 REV.

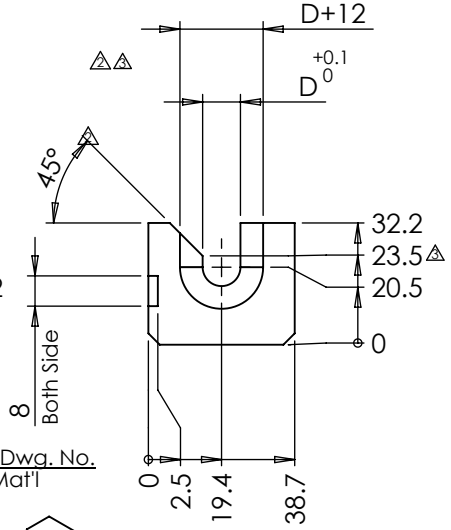
Rt=12

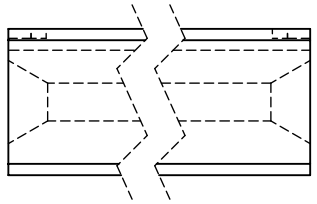
✓	Mat'l Tick by user
	Pom white
	PP Gray
	Peek Gray

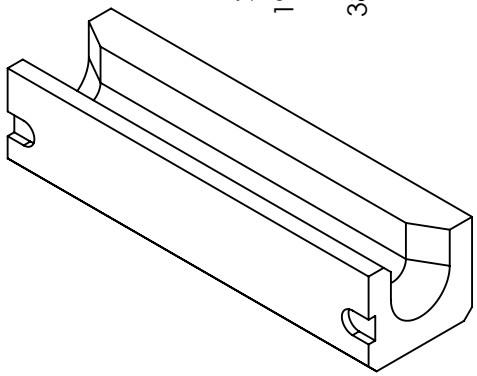












Item	D +0.1 0	For Mat'l	Check	Item	D +0.1 0	For Mat'l	Check
-01	∅10	∅8.51 to ∅9.5	Preecha	-16	∅20.5	∅20	Preecha
-02	∅12	∅10.51 to ∅11.5	Preecha	-17	∅18.5	∅18	Preecha
-03	∅7	∅5.51 to ∅6.5	Preecha	-18	∅4.4	∅4.4	Preecha RQ.304/26
-04	∅11	∅9.51 to ∅10.5	Preecha	-19			
-05	∅14	∅13 [4]	Preecha	-20			
-06	∅8	∅6.51 to ∅7.5	RQ. 2875	-21			
-07	∅17	∅16	For PMA.	-22			
-08	∅9	∅7.51 to ∅8.5	RQ.2969	-23			
-09	∅6	∅5.01 to ∅5.5	RQ.118/14	-23			
-10	∅6.5	∅6.1 to ∅6.2	RQ.773/14	-24			
-11	∅15	∅14 to ∅14.8	RQ.263/16	-26			
-12	∅13	∅11.6 to ∅12.8	RQ.271/16	-27			
-13	∅16	∅15 to ∅15.8	RQ346/16	-28			
-14	∅14	∅13.2	RQ346/16	-29			
-15	∅18	∅17	RQ.507/16	-30			

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Engineering Design center
G/M.1 & M/A. Div.

Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1

Use for: Pusher

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE
2	----	22 Feb'13	Preecha	Change: 20° ==> 45° etc.; Rq. 2836
3	----	21 Oct'13	Preecha	Add: 23.5, Delete 4 etc.; Rq. 3276
4	----	3 Aug'16	Preecha	Change: 18 ==> 17 etc.;
5	----	7 May' 20	Preecha	Add: Item Mat'l Rq.174/20
6	----	22 May' 20	Preecha	Add: 6

UNLESS SPECIFIED C0.1 OR R0.1
GENERAL TOLERANCE
JISB0405 MACHING WORKING

LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S 6.3S 25S 100S ~ UNCUT

M/C. BM-165 | Q'ty 4
MATERIAL See table [5]
FINISH -
HEAT TREATMENT -

List Request No. -

3 RD ANGLE

SCALE : | SHEET / | UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Chonknok	Preecha	Preecha	Preecha

DATE 21 Mar' 11

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

TITLE: G-Channel (3)

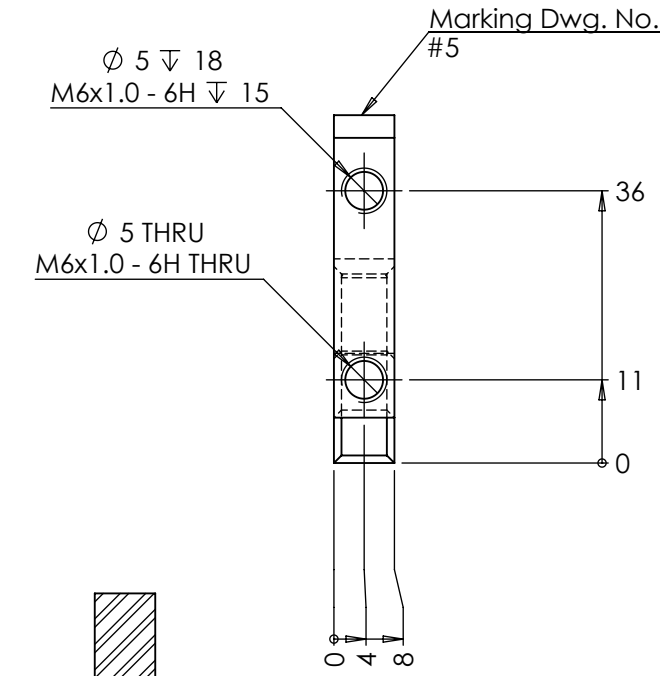
PART No. TGM-12-390-5-xx

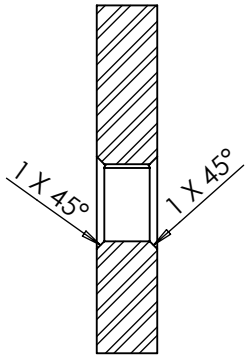
6 REV.

PART No. TGM-12-390-8-xx △3

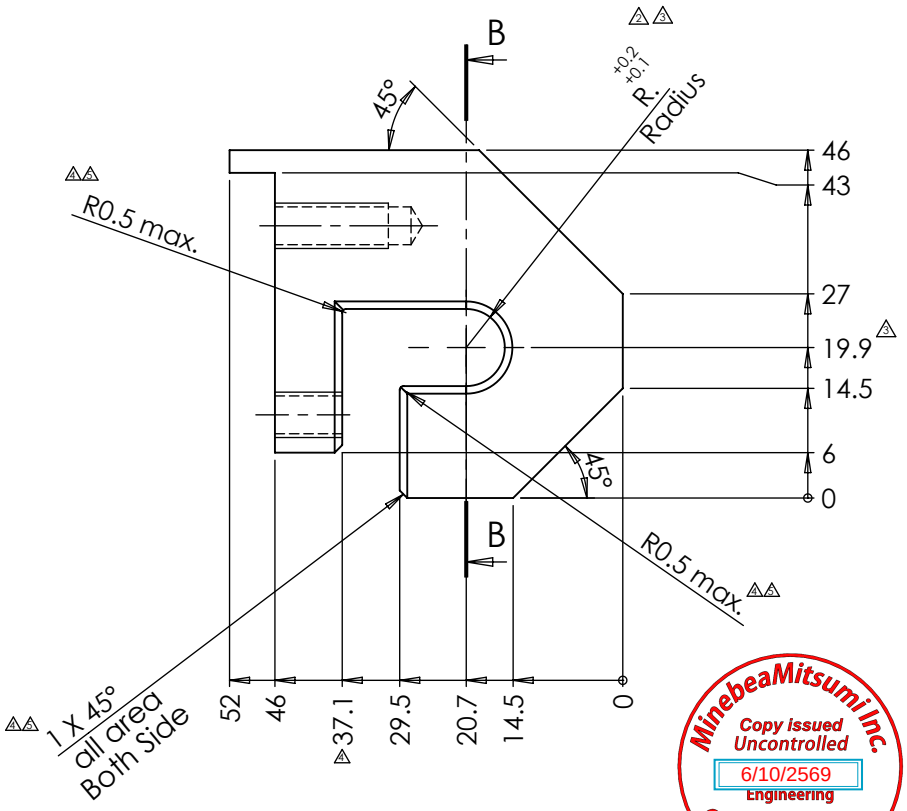
5 REV.

Rt=6





SECTION B-B
SCALE 1 : 1



MinebeaMitsumi Inc.
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6/10/2569
Engineering Design center
G/M.1 & M/A. Div.

Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1

Use for: Pusher

Item	△2△3 +0.2 +0.1 R	For Mat'l	Check
-01	5.1	∅8.51 to ∅9.5	Preecha
-02	6	∅10.51 to ∅11.5	Preecha
-03	4.5	∅6.51 to ∅8.0	Preecha
-04	8	∅16	For PMA.
-05	3.8	∅6.51 to ∅7.5	RQ. 2969
-06	3	∅5.01 to ∅5.5	RQ. 118/14
-07	3.9	∅6.1 to ∅6.2	RQ. 773/14
-08	7.5	∅14	RQ. 263/16
-09	8.5	∅15	RQ. 346/16
-10	7	∅13.5	RQ. 346/16
-11	6.5	∅11.6 to ∅12.8	RQ. 469/16
-12	9	∅17	RQ507/16
-13	5.5	9.51 to ∅10.5	RQ507/16
-14	10.25	∅20	RQ121/17
-15	9.25	∅18	RQ131/21

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE
2	----	22 Jul'11	Preecha	Change: R5.1 ==> R Item etc.;
3	----	6 Feb'12	Preecha	Add: +0.2 +0.1 etc.;
4	----	8 Aug'13	Preecha	Change: R3 ==> R0.5 max. 36.1 ==> 37.1 RQ.3148 etc.;
5	----	7 Dec'13	Preecha	Change: C0.5 ==> C1 RQ.3387 etc.;

UNLESS SPECIFIED C0.1 OR R0.1	
GENERAL TOLERANCE	
JISB0405 MACHING WORKING	
LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S 6.3S 25S 100S ~ UNCUT

M/C. BM-165 | Q'ty 2

MATERIAL SS

FINISH Black Oxide

HEAT TREATMENT -

List Request No. -

3 RD ANGLE

SCALE : | SHEET / | UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Chonknok	Preecha	Preecha	Preecha

DATE 21 Mar' 11

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

TITLE: Finger

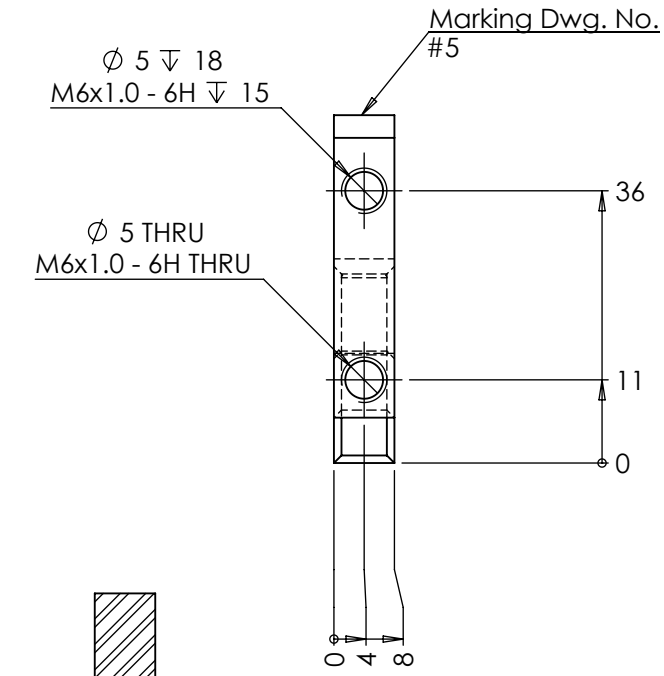
PART No. TGM-12-390-8-xx △3

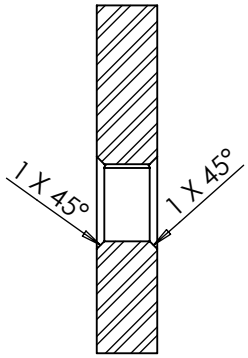
5 REV.

PART No. TGM-12-390-8-xx △3

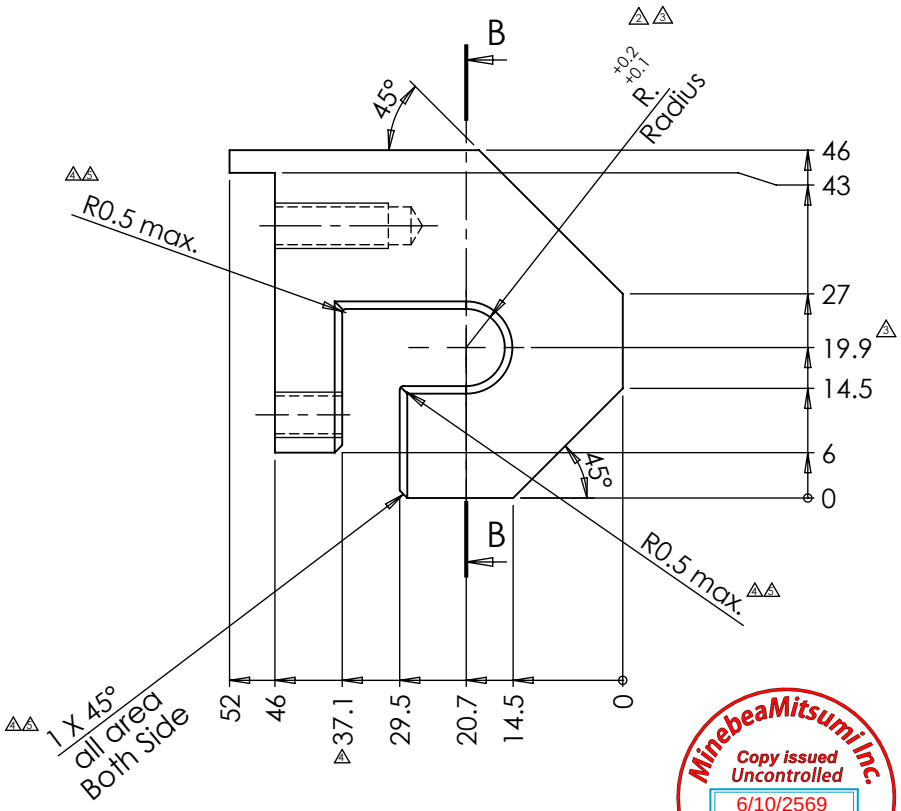
5 REV.

Rt=6





SECTION B-B
SCALE 1 : 1



MinebeaMitsumi Inc. Copy issued Uncontrolled 6/10/2569 Engineering Design center G/M.1 & M/A. Div.

Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1

Use for: Pusher

Item	△2△3 +0.2 +0.1 R	For Mat'l	Check
-16	4.4	∅4.4	Preecha RE.304/26
-17			
-18			
-19			
-20			
-21			
-22			
-23			
-24			
-25			
-26			
-27			
-28			
-29			
-30			

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE
2	----	22 Jul'11	Preecha	Change: R5.1 ==> R Item etc.;
3	----	6 Feb'12	Preecha	Add: +0.2 +0.1 etc.;
4	----	8 Aug'13	Preecha	Change: R3 ==> R0.5 max. 36.1 ==> 37.1 RQ.3148 etc.;
5	----	7 Dec'13	Preecha	Change: C0.5 ==> C1 RQ.3387 etc.;

UNLESS SPECIFIED C0.1 OR R0.1	
GENERAL TOLERANCE	
JISB0405 MACHING WORKING	
LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S 6.3S 25S 100S ~ UNCUT

M/C. BM-165 | Q'ty 2

MATERIAL SS

FINISH Black Oxide

HEAT TREATMENT -

List Request No. -

3 RD ANGLE

SCALE : | SHEET / | UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Chonknok	Preecha	Preecha	Preecha

DATE 21 Mar' 11

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

TITLE: Finger

PART No. TGM-12-390-8-xx △3

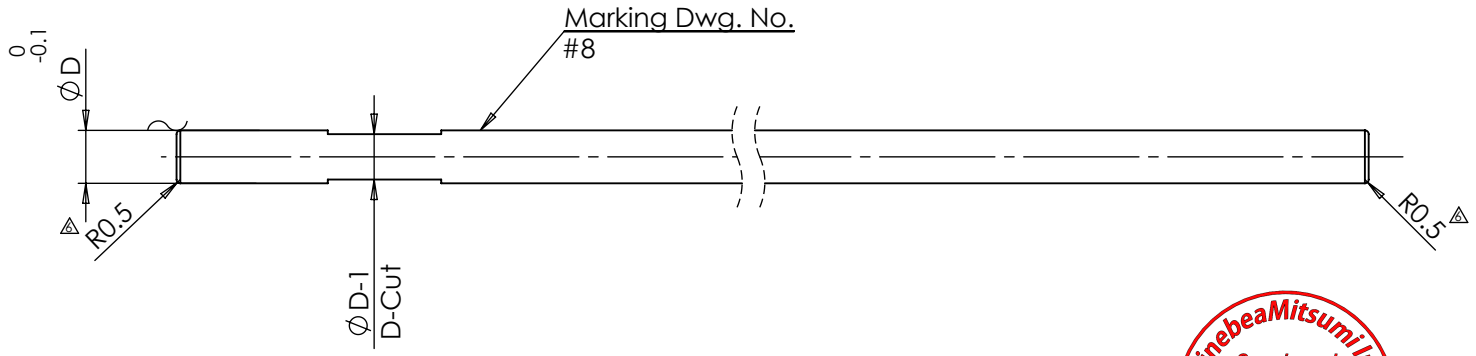
5 REV.

PART No.
TGM-12-390-12B-xx

8 REV.

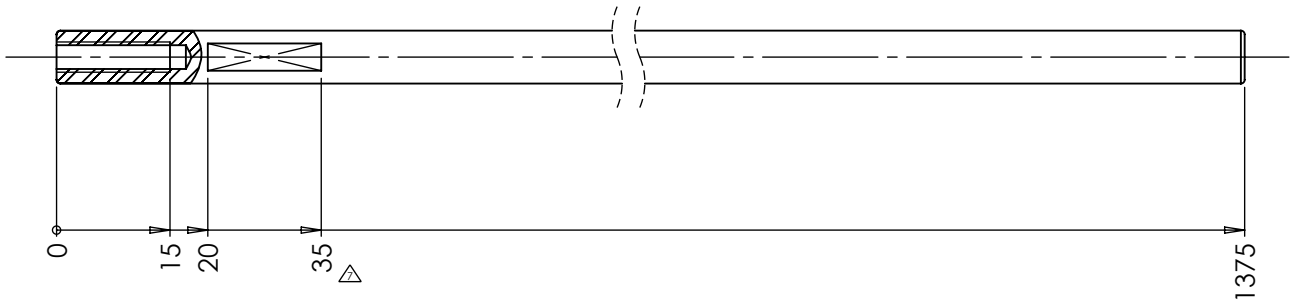
△3 Formula: ∅D = OD. Mat'l - 0.5

Rt=12 (~)



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Design center
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△3△2 Screw = M
Left Hand
See table



Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1

△3 Use for: Model Pusher

Item	∅D	M Screw Left △3	Mat'l	Check
-01	10	M6x1 △5	11.2	Preecha
-02	8	M5x0.8	9	Preecha
-03	5	M4x0.7	5.5	Preecha
-04	6	M4x0.7	6.51	Preecha
-05	15.5 △4	M8x1.25 △4	16	Preecha
-06	7	M4x0.7	7.7	RQ. 2969
-07	4.8	M4x0.7	5.12	RQ. 118/14
-08	6.2	M4x0.7	6.3	RQ. 471/14
-09	6.3 △3	M4x0.7	6.8	RQ. 459/14
-10	6.8	M4x0.7	7.1	RQ. 482/14
-11	9	M5x0.8	∅9.2 to ∅9.6	RQ. 601/14
-12	5.6	M4x0.7	∅6.1 to ∅6.2	RQ. 774/14
-13	△8 13.5	M8x1.25 △4	14	RQ.263/16
-14	△8 11.5	M8x1.25 △5	12	RQ.271/16
-15	△8 14.5	M8x1.25 △4	15	RQ.346/16

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE
8	----	9 Mar'20	Preecha	Change: Item -13 to-16 Change spec. gap = 0.5
3	----	9 Oct'14	Preecha	Change: Item -05 16 ==> 15.5 etc.; Rq. 601/14
4	----	1 Sep'16	Preecha	Change: M5x0.8 ==> M6x1, M8x1.25 etc.;
5	----	20 Oct'16	Preecha	Change: M5x0.8 ==> M6x1, M8x1.25 etc.; Item -01, -14
6	----	20 Dec'16	Preecha	Change: R1 ==> R0.5 Rq. 646/16
7	----	2 Dec'17	Preecha	Change: 25 ==> 35 Rq. 602/17

UNLESS SPECIFIED C0.1 OR R0.1	
GENERAL TOLERANCE	
JISB0405 MACHING WORKING	
LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S ▽ 6.3S ▽ 25S ▽ 100S ▽ ~ UNCUT

M/C. BM-165 | Q'ty 1

MATERIAL SUS 304

FINISH -

HEAT TREATMENT -

List Request No. 2938

3 RD ANGLE

SCALE : | SHEET / | UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Chonknok	Preecha	Preecha	Preecha

DATE 25 May' 13

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

TITLE: Shaft Pusher Mat'l (B)

PART No. TGM-12-390-12B-xx

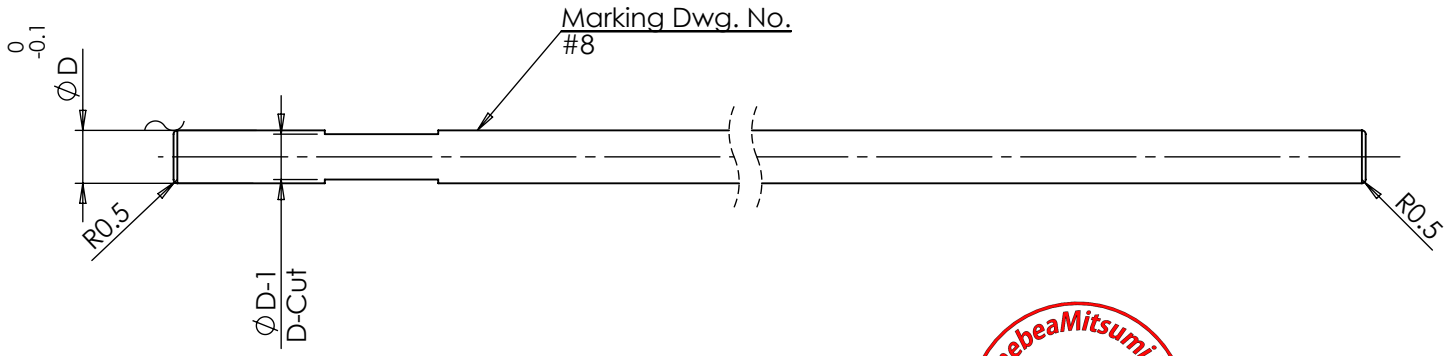
8 REV.

PART No. TGM-12-390-12B-xx

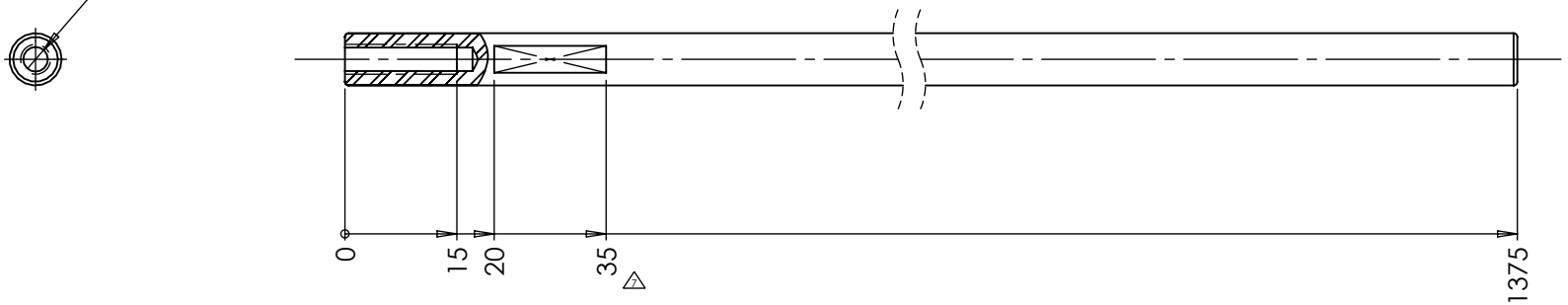
8 REV.

△3 Formula: ∅D = OD. Mat'l - 0.5

Rt=12 (~)



△3△2 Screw = M
Left Hand
See table



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Item	∅D	△3 M Screw Left	Mat'l	Check
-16	△8 12.7	M8x1.25	△3 13.2	Preecha
-17	16.5	M8x1.25	17	RQ.705/16
-18	19.5	M8x1.25	20	RQ.121/17
-19	17.5	M8x1.25	18	RQ.13/12
-20	4	M3x0.5	4.4	Preecha RQ.304/26
-21				
-22				
-23				
-24				
-25				
-26				
-27				
-28				
-29				
-30				

Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1

△3 Use for: Model Pusher

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE
8	----	9 Mar'20	Preecha	Change: Item -13 to-16 Change spec. gap = 0.5
3	----	9 Oct'14	Preecha	Change: Item -05 16 ==> 15.5 etc.; Rq. 601/14
4	----	1 Sep'16	Preecha	Change: M5x0.8 ==> M6x1, M8x1.25 etc.;
5	----	20 Oct'16	Preecha	Change: M5x0.8 ==> M6x1, M8x1.25 etc.; Item -01, -14
6	----	20 Dec'16	Preecha	Change: R1 ==> R0.5 Rq. 646/16
7	----	2 Dec'17	Preecha	Change: 25 ==> 35 Rq. 602/17

UNLESS SPECIFIED C0.1 OR R0.1	
GENERAL TOLERANCE	
JISB0405 MACHING WORKING	
LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S 6.3S 25S 100S ~ UNCUT

M/C. BM-165 | Q'ty 1

MATERIAL SUS 304

FINISH -

HEAT TREATMENT -

List Request No. 2938

3 RD ANGLE

SCALE : | SHEET / | UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Chonknok	Preecha	Preecha	Preecha

DATE 25 May' 13

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

TITLE: Shaft Pusher Mat'l (B)

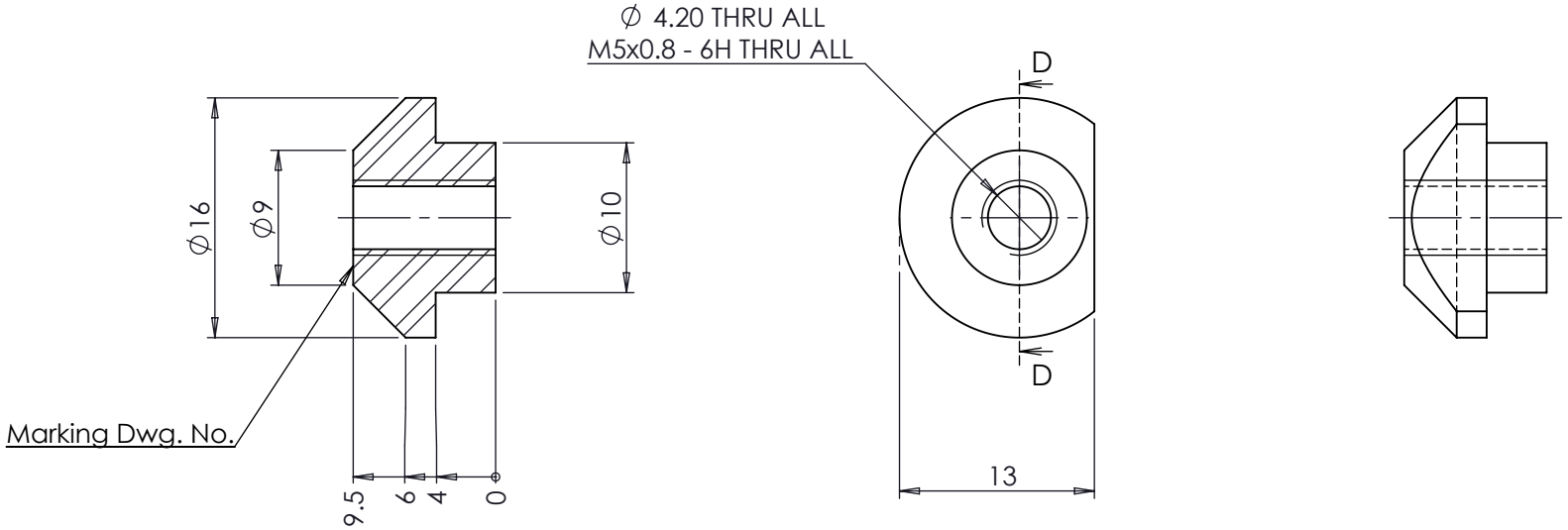
PART No. TGM-12-390-12B-xx

8 REV.

PART No.
TGM-12-390-13

1 REV.

Rt=6



SECTION D-D
SCALE 2 : 1

Note: 1. Break all Corner approx by C0.1
2. General Tolerance ±0.1

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REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE

UNLESS SPECIFIED C0.1 OR R0.1	
GENERAL TOLERANCE	
JISB0405 MACHING WORKING	
LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S 6.3S 25S 100S UNCUT

M/C. BM-165 | Q'ty 1
MATERIAL SUS 304
FINISH -
HEAT TREATMENT -

List Request No. -
3 RD ANGLE
SCALE : | SHEET / | UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Chonknok	Preecha	Preecha	Preecha

DATE 20 Oct' 14

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

TITLE: Nut lock plate

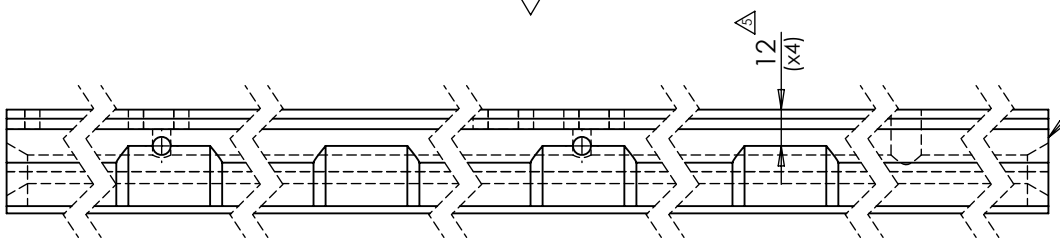
PART No. TGM-12-390-13

1 REV.

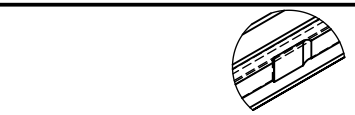
PART No. TGM-12-390-20-xx

9 REV.

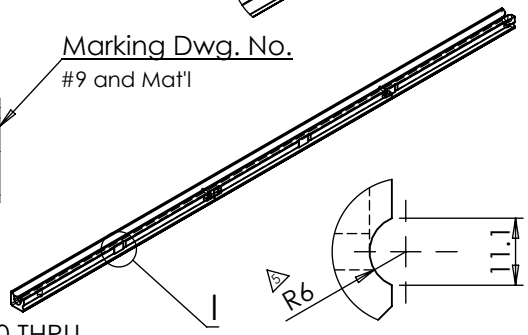
Rt=12

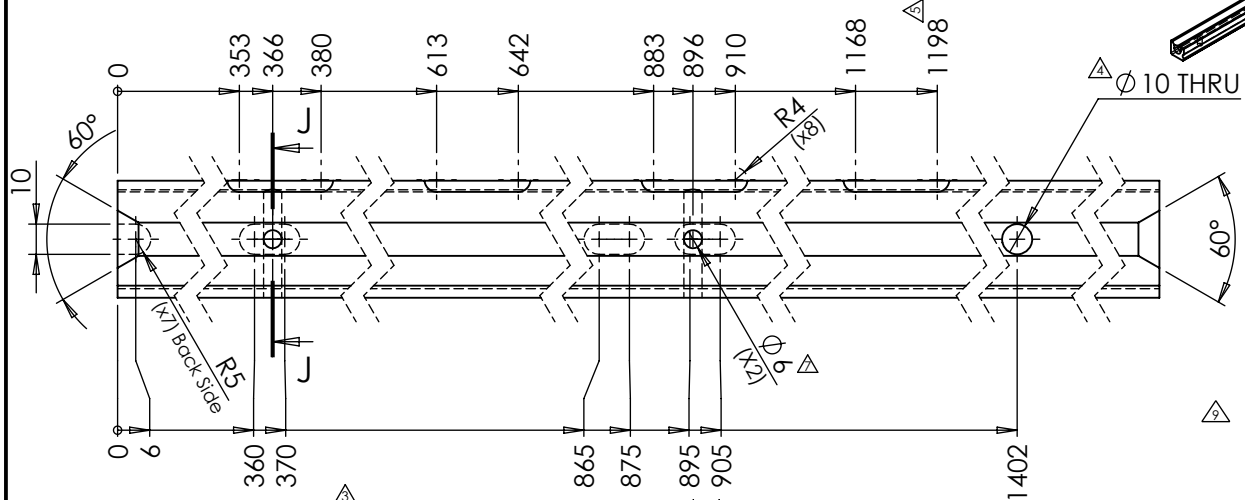


Marking Dwg. No.
#9 and Mat'l

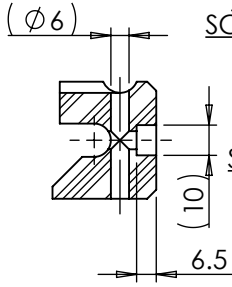


DETAIL I
SCALE 1 : 5

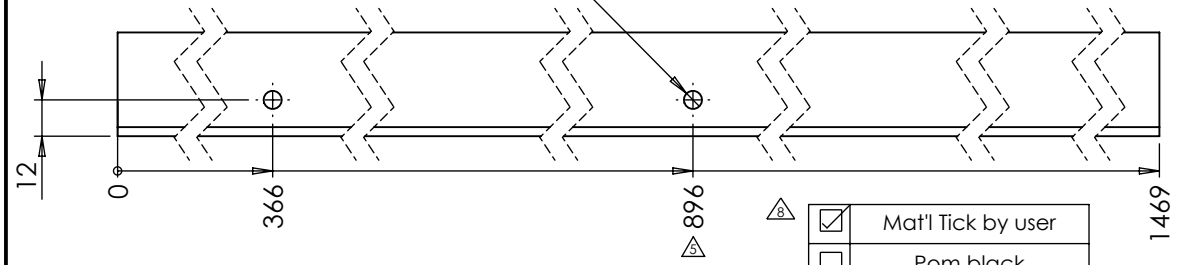


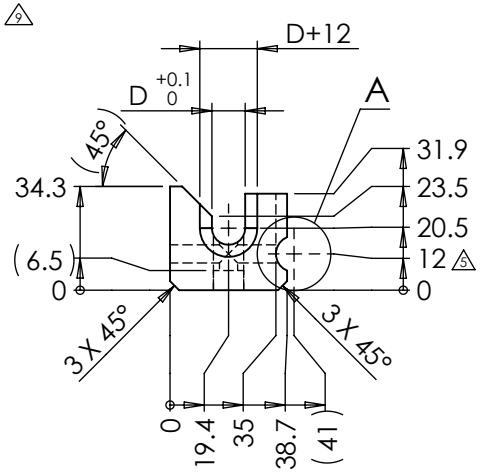


DETAIL A
SCALE 2 : 2.5



SECTION J-J
SCALE 1 : 2.5





☑	Mat'l Tick by user
☐	Pom black
☐	PP Gray
☐	Peek Gray

Use for : Part Torn

Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1
3. Accept Body Bend 0.5 to 2 mm.

Item	ØD	for Mat'l	Check
-01	Ø6	Ø5.01 to Ø5.5	Preecha
-02	Ø7	Ø5.51 to Ø6.5	Preecha
-03	Ø8	Ø6.51 to Ø7.5	Preecha
-04	Ø9	Ø7.51 to Ø8.5	Preecha
-05	Ø10	Ø8.51 to Ø9.5	Preecha
-06	Ø11	Ø9.51 to Ø10.5	Preecha
-07	Ø12	Ø10.5 to Ø11.5	Preecha
-08	Ø18	Ø16	Preecha
-09	Ø15	Ø14	Preecha
-10	Ø13	Ø11.6 to Ø12.8	Preecha
-11	Ø16	Ø15	Preecha RQ.346/16
-12	Ø14	Ø13.2	Preecha RQ.346/16
-13	Ø18	Ø17	Preecha RQ.507/16
-14	Ø20.5	Ø20	Preecha
-15	Ø18.5	Ø18	Preecha RQ.13/21
-16	Ø4.4	Ø4.4	Preecha RQ.304/26
-17			
-18			
-19			
-20			
-21			
-22			
-23			
-24			
-25			

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REV.	ECN No.	DATE	NAME	DESCRIPTION
8	----	7 May' 20	Preecha	Add: Item Mat'l Rq.174/20
9	----	15 Feb' 22	Preecha	Add : Section J-J etc,.
3	----	19 Dec' 14	Preecha	Change : Add : Bore 4x10 Depth 6.5 etc;.
4	----	6 Feb' 15	Preecha	Change : 4x10 Depth 6.5 R5 Slide etc;.
5	----	2 Apr' 16	Preecha	Add : Oil Key etc;. Rq : 143/16
6	----	3 Aug' 16	Preecha	Change : Ø18 ==> 17 etc;.
7	----	24 Aug' 16	Preecha	Change : ØD-1 ==> Ø6 etc;.

UNLESS SPECIFIED C0.1 OR R0.1
GENERAL TOLERANCE
JISB0405 MACHING WORKING

LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S 6.3S 25S 100S ~ UNCUT

M/C. BM165
Q'ty 1
MATERIAL 8 See table
FINISH -
HEAT TREATMENT -

List Request No. 708/14
3 RD ANGLE
SCALE : SHEET / UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Chonknok	Chonknok	Preecha	Preecha

DATE 25 Nov' 14

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

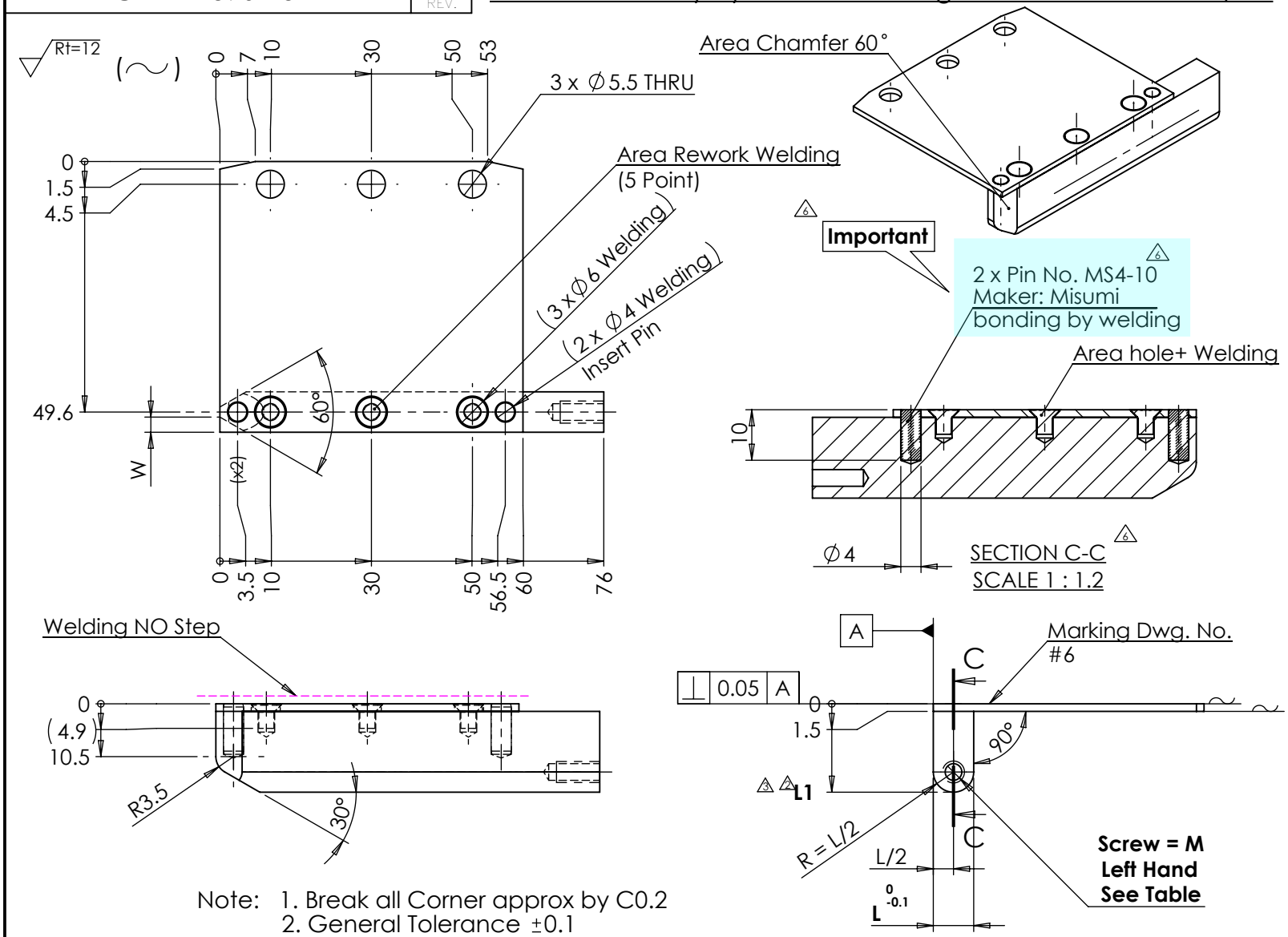
TITLE: G-Channel (3 In 1)

PART No. TGM-12-390-20-xx

9 REV.

PART No. TGM-12-390-25-xx | 6 REV.

For: Vendor Ass'y by Process Welding Close Hole Send to G/M1



Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1

Item	L	L1	W	M Screw Left	Use For Mat'l	Check
-16	17.5	22	6	M8x1.25	∅18	Preecha
-17	4	15.8	2	M3x0.5	∅4.4	Preecha RQ.304/26
-18						
-19						
-20						
-21						
-22						
-23						
-24						
-25						
-26						
-27						
-28						
-29						
-30						

MinebeaMitsumi Inc.
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6/10/2569
Engineering Design center
G/M.1 & M/A. Div.

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	----	ENGINEERING RELEASE
2	----	16 Jul'16	Preecha	Change: 17.5 ===> 21.5 Rq. 346/16
3	----	31 Aug'16	Preecha	Change: M5x0.8 ===> M8x1.25 etc.: Rq. 346/16
4	----	20 Oct'16	Preecha	Change: M5x0.8 ===> M8x1.25 etc.: Rq. 346/16
5	----	16 Jun'17	Preecha	Change: M5x0.8 ==> M8x1.25 etc.; Item -08, -11
6	----	3 Feb'26	Preecha	Add: Section etc.; RQ.37/26

UNLESS SPECIFIED C0.1 OR R0.1
GENERAL TOLERANCE
JISB0405 MACHING WORKING

LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S | 6.3S | 25S | 100S | ~ UNCUT

M/C. BM-165 | Q'ty 1
MATERIAL SUS 304
FINISH -
HEAT TREATMENT -

List Request No. 60/15
3 RD ANGLE
SCALE : | SHEET 1/2 | UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Phalachai	Aphisek	Preecha	Preecha

DATE 13 Feb' 15

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

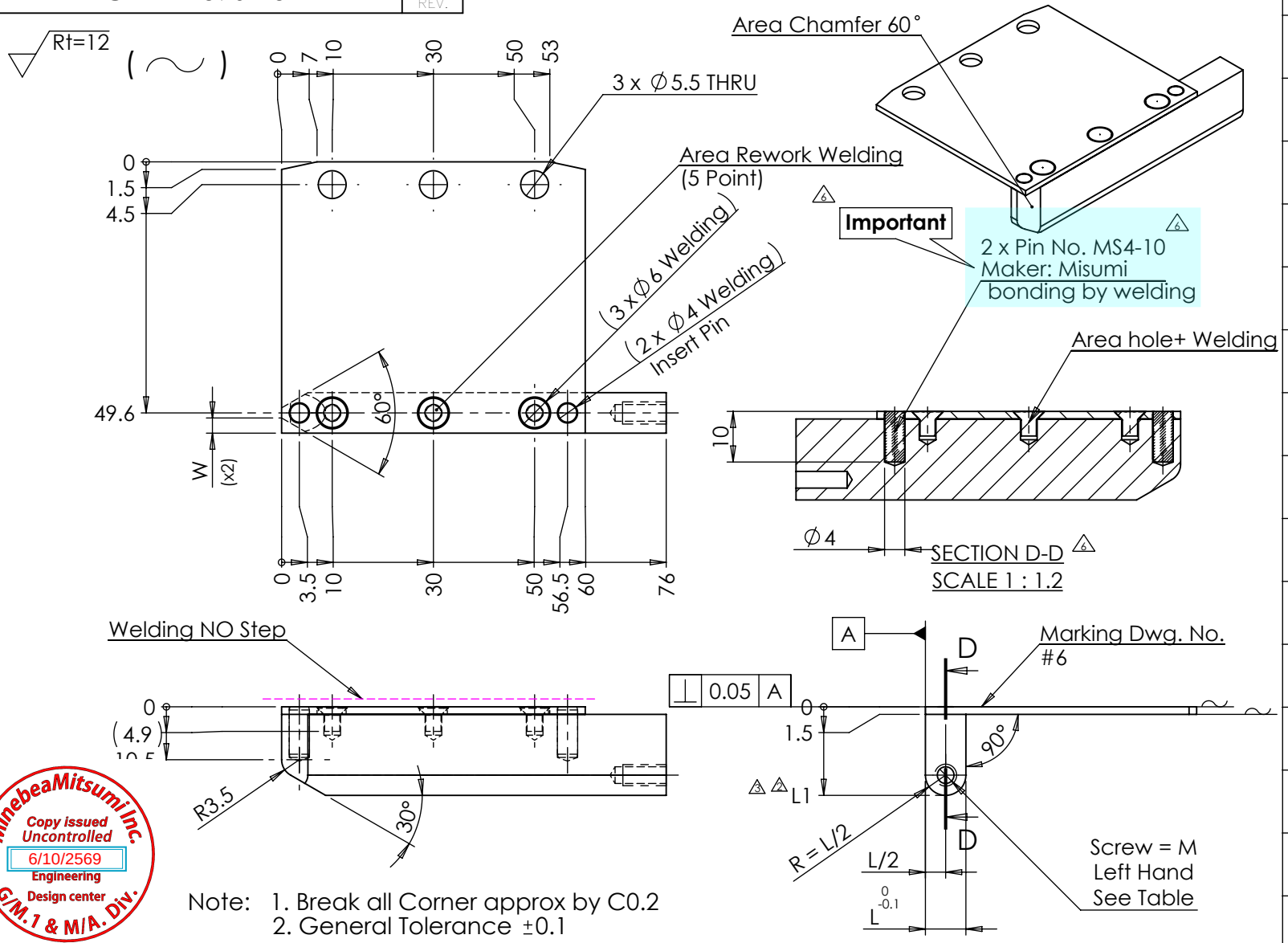
TITLE: Plate Pusher Mat'l (2) 6

PART No. TGM-12-390-25-xx | 6 REV.

PART No. TGM-12-390-25-xx

6 REV.

For: Vendor Ass'y by Process Welding Close Hole Send to G/M1



Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1

Item	L	L1	W	M Screw Left	Use For Mat'l	Check
-01	5	15.7	1	M4x0.7	∅5.01 to ∅5.5	Preecha
-02	5.6	15.95	2	M4x0.7	∅6.1 to ∅6.2	Preecha
-03	6	16.2	1.3	M4x0.7	∅6.3	Preecha
-04	6.5	16.7	2	M4x0.7	∅6.51 to ∅7.5	Preecha
-05	7.5	16.7	3	M4x0.7	∅7.51 to ∅8.5	Preecha
-06	8	17.2	3	M5x0.8	∅8.5 to ∅8.8	Preecha
-07	9	17.7	3	M5x0.8	∅9.2 to ∅9.6	Preecha
-08	10	18.7	3	M6x1	∅10.5 to ∅11.5	Preecha
-09	15.5	21.2	6	M8x1.25	∅16	Preecha
-10	14	20.2	6	M8x1.25	∅14	Preecha
-11	12	19.7	3	M8x1.25	∅11.6 to ∅12.8	Preecha
-12	14.5	20.7	6	M8x1.25	∅15	Preecha
-13	12.7	19.7	6	M8x1.25	∅13.2	Preecha
-14	16.5	22.25	6	M8x1.25	∅17	RQ.507/16
-15	19.5	22	6	M8x1.25	∅20	RQ.121/17

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REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	----	ENGINEERING RELEASE
2	----	16 Jul'16	Preecha	Change: 17.5 ===> 21.5 Rq. 346/16
3	----	31 Aug'16	Preecha	Change: M5x0.8 ===> M8x1.25 etc.: Rq. 346/16
4	----	20 Oct'16	Preecha	Change: M5x0.8 ===> M8x1.25 etc.: Rq. 346/16
5	----	16 Jun'17	Preecha	Change: M5x0.8 ==> M8x1.25 etc.; Item -08, -11
6	----	3 Feb'26	Preecha	Add: Section etc.; RQ.37/26

UNLESS SPECIFIED C0.1 OR R0.1	
GENERAL TOLERANCE	
JISB0405 MACHING WORKING	
LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S 6.3S 25S 100S ~ UNCUT

M/C. BM-165 | Q'ty 1

MATERIAL SUS 304

FINISH -

HEAT TREATMENT -

List Request No. 60/15

3 RD ANGLE

SCALE : | SHEET / | UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Phalachai	Aphisek	Preecha	Preecha

DATE 13 Feb' 15

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

TITLE: Plate Pusher Mat'l (2)

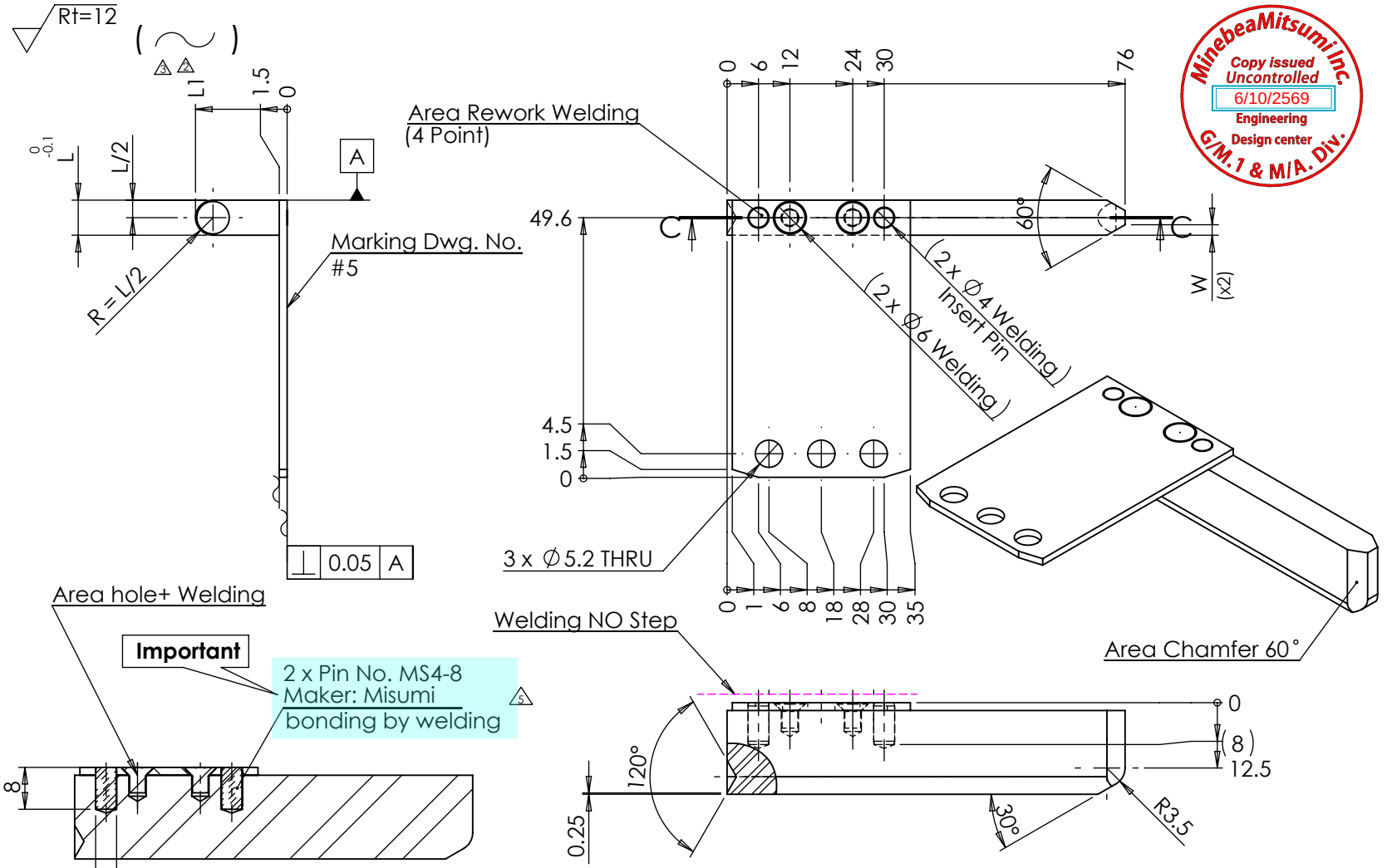
PART No. TGM-12-390-25-xx

6 REV.

PART No. TGM-12-390-26-xx

5 REV.

For: Vendor Ass'y by Process Welding Close Hole Send to G/M1 △4



SECTION C-C △5
SCALE 1 : 1.2

Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1

Item	L	L1	W	Use For Mat'l	Check
-16	17.5	22	6	Ø18	Preecha
-17	4	15.8	2	Ø4.4	Preecha RQ.304/26
-18					
-19					
-20					
-21					
-22					
-23					
-24					
-25					
-26					
-27					
-28					
-29					
-30					

MinebeaMitsumi Inc. Copy issued Uncontrolled 6/10/2569 Engineering Design center G/M.1 & M/A. Div.

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE
2	----	16 Jul'16	Preecha	Change: 17.5 ===> 21.5 Rq. 346/16
3	----	31 Aug'16	Preecha	Change: 21.5 ===> L1 etc.: Rq. 346/16
4	----	16 Jun'17	Preecha	Cancle Item Screw7 4
5	----	3 Feb'26	Preecha	Add: Section etc.; RQ.37/26

UNLESS SPECIFIED C0.1 OR R0.1
GENERAL TOLERANCE
JISB0405 MACHING WORKING

LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S 6.3S 25S 100S UNCUT

M/C. BM-165 | Q'ty 1
MATERIAL SUS 304
FINISH -
HEAT TREATMENT -

List Request No. 60/15
3 RD ANGLE
SCALE : | SHEET / | UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Phalachai	Aphisek	Preecha	Preecha

DATE 13 Feb' 15

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

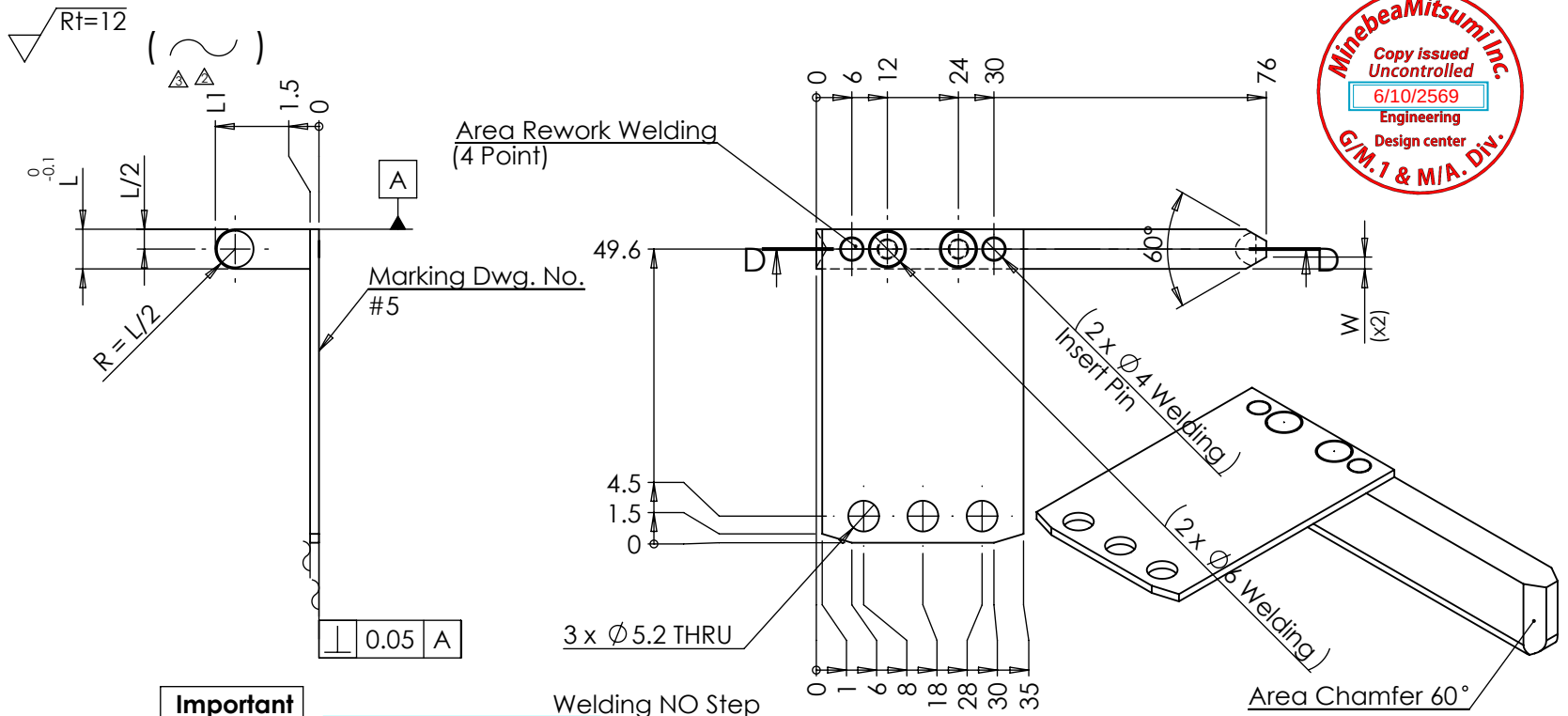
TITLE: Plate Pusher Mat'l (1)

PART No. TGM-12-390-26-xx

5 REV.

PART No. TGM-12-390-26-xx | 5 REV.

For: Vendor Ass'y by Process Welding Close Hole Send to G/M1 △4

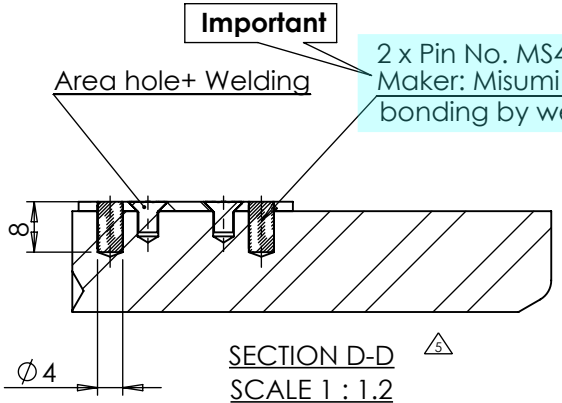


MinebeaMitsumi Inc.
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6/10/2569
Engineering Design center
G/M.1 & M/A. Div.

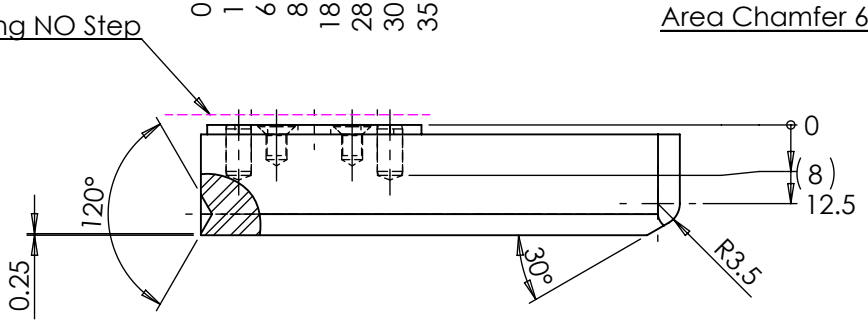
Item	L	L1	W	Use For Mat'l	Check
-01	5	15.7	1	Ø5.01 to Ø5.5	Preecha
-02	5.6	15.95	2	Ø6.1 to Ø6.2	Preecha
-03	6	16.2	1.3	Ø6.3	Preecha
-04	6.5	16.7	2	Ø6.51 to Ø7.5	Preecha
-05	7.5	16.7	3	Ø7.51 to Ø8.5	Preecha
-06	8	17.2	3	Ø8.5 to Ø8.8	Preecha
-07	9	17.7	3	Ø9.2 to Ø9.6	Preecha
-08	10	18.7	3	Ø10.5 to Ø11.5	Preecha
-09	15.5	21.2	6	Ø16	Preecha
-10	14	20.2	6	Ø14	Preecha
-11	12	19.7	3	Ø11.6 to Ø12.8	Preecha
-12	14.5	20.7	6	Ø15	Preecha
-13	12.7	19.7	6	Ø13.2	Preecha
-14	16.5	22.25	6	Ø17	RQ.507/16
-15	19.5	22	6	Ø20	RQ.121/17

Important

2 x Pin No. MS4-8
Maker: Misumi
bonding by welding



SECTION D-D △5
SCALE 1 : 1.2



Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE
2	----	16 Jul'16	Preecha	Change: 17.5 ===> 21.5 Rq. 346/16
3	----	31 Aug'16	Preecha	Change: 21.5 ===> L1 etc.: Rq. 346/16
4	----	16 Jun'17	Preecha	Cancle Item Screw7 4
5	----	3 Feb'26	Preecha	Add: Section etc.; RQ.37/26

UNLESS SPECIFIED C0.1 OR R0.1
GENERAL TOLERANCE
JISB0405 MACHING WORKING

LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S 6.3S 25S 100S ~ UNCUT

M/C. BM-165 | Q'ty 1
MATERIAL SUS 304
FINISH -
HEAT TREATMENT -

List Request No. 60/15
3 RD ANGLE
SCALE : | SHEET / | UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Phalachai	Aphisek	Preecha	Preecha

DATE 13 Feb' 15

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

TITLE: Plate Pusher Mat'l (1)

PART No. TGM-12-390-26-xx | 5 REV.